EE3621: Digital Signal Processing | III B.Tech EEE

Faculty Reference Document — Textbook Replacement

PREFACE FOR FACULTY

In real-time DSP, input sequences $x[n]$ (speech, audio, radar returns) are effectively infinite in length, while FIR filter impulse responses $h[n]$ have finite length M. Computing the entire convolution at once is impossible due to memory limits and unacceptable latency. The **Overlap-Add (OLA)** method segments the input into non-overlapping blocks of length L and performs block convolution via FFT/IFFT.

Pedagogical Strategy:

1. Formulate input signal decomposition into disjoint blocks: $x[n] = \sum_{m=0}^{\infty} x_m[n - mL]$.
2. Establish block zero-padding requirement: $N \geq L + M - 1$.
3. Compute frequency-domain circular convolution for each block: $y_m[n] = \mathrm{IDFT}_N \mathrm{DFT}_N x_m \cdot \mathrm{DFT}_N h$.
4. Synthesize the total output by adding the overlapping $M - 1$ tails between consecutive blocks.
5. Derive the optimal block length L_{opt} that minimizes total computational cost per output sample.

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1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Partition** long streaming sequences into blocks suitable for Overlap-Add filtering.
2. **Execute** complete numerical block filtering using the Overlap-Add method.
3. **Determine** the required FFT size N to prevent aliasing within blocks.
4. **Calculate** overall throughput and latency for real-time DSP implementations.

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2. MATHEMATICAL FOUNDATIONS

2.1 Overlap-Add Block Decomposition

Let $x[n]$ be an infinite input sequence and $h[n]$ an FIR filter of length M. Segment $x[n]$ into non-overlapping blocks of length L:

$$x_m[n] = \{x[n+mL], \quad 0 \leq n \leq L-1 \, 0, \quad \text{otherwise}$$

Total input: $x[n] = \sum_{m=0}^{\infty} x_m[n-mL]$.

2.2 Linear Convolution of Sub-Blocks

By Linearity of LTI systems:

$$y[n] = x[n] * h[n] = \left(\sum_{m=0}^{\infty} x_m[n-mL] \right) * h[n] = \sum_{m=0}^{\infty} (x_m[n-mL] * h[n]) = \sum_{m=0}^{\infty} y_m[n-mL]$$

Where each block convolution $y_m[n] = x_m[n] * h[n]$ has length:

$$N = L + M - 1$$

2.3 FFT Implementation & Overlap-Addition

1. Choose FFT size $N = 2^k \geq L + M - 1$.
2. Zero-pad $x_m[n]$ with $N - L$ zeros to length N.
3. Zero-pad $h[n]$ with $N - M$ zeros to length N.
4. Precompute $H[k] = \text{DFT}_N h[n]$.
5. For each block m:
   - $X_m[k] = \text{DFT}_N x_m[n]$
   - $Y_m[k] = X_m[k] \cdot H[k]$
   - $y_m[n] = \text{IDFT}_N Y_m[k]$
6. Reconstruct output $y[n]$: Overlap the last $M-1$ points of block m with the first $M-1$ points of block $m+1$ and add.

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3. WORKED NUMERICAL EXAMPLES

Example 13.1: Complete Numerical Overlap-Add Filtering

Problem: Filter input $x[n] = \underset{\uparrow}{1}, 2, -1, 2, 3, -2, 0, 1, 2, 1$ using FIR filter $h[n] = \underset{\uparrow}{1}, 2, 1$ ($M = 3$) using Overlap-Add with block length $L = 4$.

Solution:

- $L = 4, ; M = 3 \implies N = L + M - 1 = 4 + 3 - 1 = 6$. (Pad to $N = 6$).
- Partition input into blocks of $L = 4$:

- $x_0[n] = 1, 2, -1, 2, 0, 0$ (padded with 2 zeros)
- $x_1[n] = 3, -2, 0, 1, 0, 0$
- $x_2[n] = 2, 1, 0, 0, 0, 0$

- Filter padded: $h[n] = 1, 2, 1, 0, 0, 0$.
- **Block Convolutions ($y_m[n] = x_m * h$):**
  - $y_0[n] = 1, 2, -1, 2 * 1, 2, 1 = \underset{\uparrow}{1}, 4, 4, 1, 3, 2$
  - $y_1[n] = 3, -2, 0, 1 * 1, 2, 1 = \underset{\uparrow}{3}, 4, -1, 0, 2, 1$
  - $y_2[n] = 2, 1, 0, 0 * 1, 2, 1 = \underset{\uparrow}{2}, 5, 4, 1, 0, 0$
- **Overlap-Addition ($L = 4$ shift):**

Misplaced \hline

Result:

$$y[n] = \underset{\uparrow}{1}, 4, 4, 1, 6, 6, -1, 0, 4, 6, 4, 1$$

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4. UNIVERSITY EXAMINATION QUESTIONS & MARKING RUBRIC

Question 1 (15 Marks)

(a) Describe the Overlap-Add method for linear filtering of long data sequences. Explain why zero-padding is necessary. *(7 Marks)* **(b)** Use the Overlap-Add method to filter $x[n] = \underset{\uparrow}{1}, 2, 0, -1, 3, 1, 2, -1$ with $h[n] = \underset{\uparrow}{1}, 1, 1$ using block length $L = 4$. *(8 Marks)*

Model Answer & Step-by-Step Marking Rubric:

- **Part (a):**
  - Mathematical formulation of block decomposition and overlap additions *(4 Marks)*
  - Explanation of $N \geq L + M - 1$ to prevent circular aliasing *(3 Marks)*
- **Part (b):**
  - Block setup: $x_0 = 1, 2, 0, -1, ; x_1 = 3, 1, 2, -1$ *(2 Marks)*
  - Convolutions: $y_0 = 1, 2, 0, -1 * 1, 1, 1 = 1, 3, 3, 1, -1, -1$
    $y_1 = 3, 1, 2, -1 * 1, 1, 1 = 3, 4, 6, 2, 1, -1$ *(4 Marks)*
  - Assembly: $y[n] = 1, 3, 3, 1, (-1 + 3), (-1 + 4), 6, 2, 1, -1 = 1, 3, 3, 1, 2, 3, 6, 2, 1, -1$ *(2 Marks)*

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5. PYTHON VERIFICATION SCRIPT

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import numpy as np

x = np.array([1, 2, 0, -1, 3, 1, 2, -1])
h = np.array([1, 1, 1])
y = np.convolve(x, h)
print("Direct Linear Convolution:", y)
```